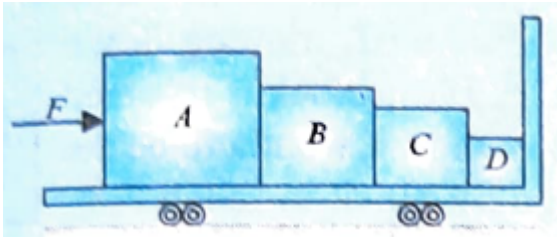
Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

A toy cart has mass of $4\ \text{kg}$ and is kept on a smooth horizontal Surface. Four blocks A, B, C and D of masses $2\ \text{kg}, 2\ \text{kg}, 1\ \text{kg}$ and $1\ \text{kg}$, respectively, have been placed on the cart. A horizontal force of $F = 60\ \text{N}$ is applied to the block A (see figure). Find the contact force between block D and the front vertical wall of the cart.



(A) 23

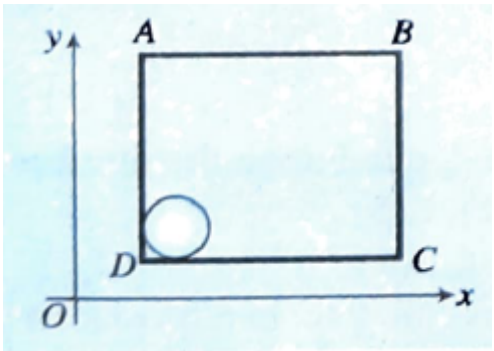
(B) 24

(C) 25

(D) 26

Q.2

A solid sphere of mass $2\ \text{kg}$ is resting inside a cube as shown in figure The cube is moving with a velocity $\vec{v} = (5t\hat{i} + 2t\hat{j})\text{ms}^{-1}$. Here t is time in seconds. All surfaces are smooth. The sphere is at rest with respect to the cube. What is the total force exerted by the sphere on the cube?



(A) 24

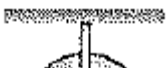
(B) 25

(C) 26

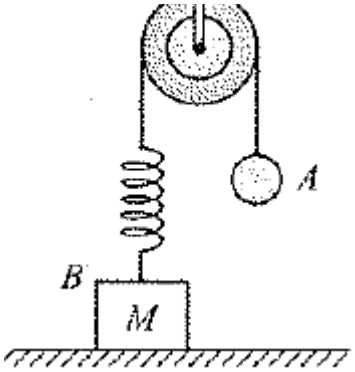
(D) 27

Q.3

In the Fig., the ball A is released from rest, when the spring is at its natural (unstretched) length. For the block B of mass M to leave contact with ground at some stage. the minimum mass of A must be



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(A) $2M$

(B) M

(C) $M/2$

(D) $M/4$

Q.4

Three forces are acting on a particle of mass m initially in equilibrium. If the first 2 forces $(R_1$ and $R_2)$ are perpendicular to each other and suddenly the third force (R_3) is removed, then the acceleration of the particle is

(A) $\frac{R_3}{m}$

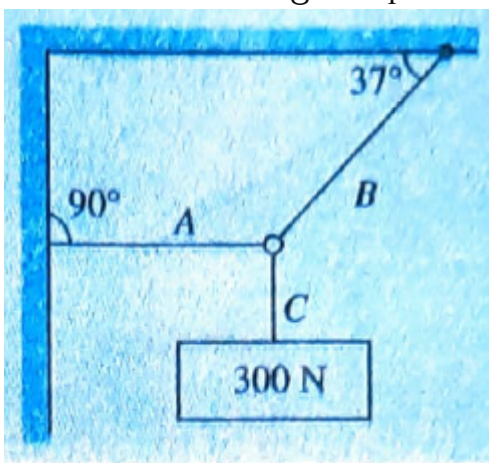
(B) $\frac{R_1+R_2}{m}$

(C) $\frac{R_1-R_2}{m}$

(D) $\frac{R_1}{m}$

Q.5

A block of mass $30\ \text{kg}$ is suspended by three strings as shown in figure. Find the tension in each string.



(A) $\text{Ta} = 450\ \text{N},\ \text{Tb} = 500\ \text{N}, \text{Tc} = 300\ \text{N}$

(B) $\text{Ta} = 400\ \text{N},\ \text{Tb} = 500\ \text{N}, \text{Tc} = 250\ \text{N}$

(C) $\text{Ta} = 400\ \text{N},\ \text{Tb} = 500\ \text{N}, \text{Tc} = 300\ \text{N}$

(D) $\text{Ta} = 400\ \text{N},\ \text{Tb} = 550\ \text{N}, \text{Tc} = 300\ \text{N}$

Q.6

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A block of mass $1\ \text{kg}$ is at rest on a horizontal table. The coefficient of static friction between the block and the table is 0.50. If $g = 10\ \text{ms}^{-2}$, then the magnitude of a force acting upwads at an angle of 60° from the horizontal that will just start the block moving is:

(A) $5\ \text{N}$

(B) $5.36\ \text{N}$

(C) $74.6\ \text{N}$

(D) $10\ \text{N}$

Q.7

A body of mass M is resting on a rough horizontal plane surface, the coefficient of friction being equal to μ. At $t = 0$ a horizontal force $F = F_0 t$ starts acting on it, where F_0 is a constant. Find the time T at which the motion Starts?

(A) $\mu Mg/F_0$

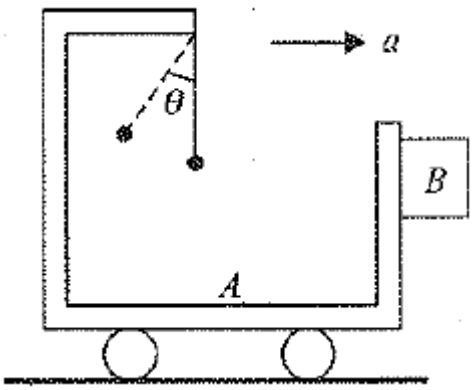
(B) $Mg/\mu F_0$

(C) $\mu F_0/Mg$

(D) None of these

Q.8

A trolley A has a simple pendulum suspended from a frame fixed to its desk. A block B is in contact on its vertical slide. The trolley is on horizontal rails and accelerates towards the right such that the block is just prevented from falling. The value of coefficient of friction between A and B is 0.5 (see Fig.). The inclination of the pendulum to the vertical is



(A) $\tan^{-1-1}\left(\frac{1}{2}\right)$

(B) $\tan^{-1}(3)$

(C) $\tan^{-1}(\sqrt{2})$

(D) $\tan^{-1}(2)$

Q.9

Starting from rest, a body slides down a 45° inclined plane in twice the time it takes to slide the same distance in the absence of friction. What is the coefficient of friction between the body and the inclined plane?

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(A) $\sqrt{3}/2$

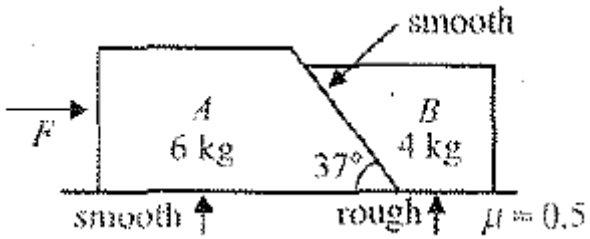
(B) $3/4$

(C) $1/2$

(D) $1/4$

Q.10

Two blocks A of $6\ \text{kg}$ and B of $4\ \text{kg}$ are placed in contact with each other as shown in Fig. There is no friction between A and ground and between both the blocks. Coefficient of friction between B and ground is 0.5. A horizontal force F is applied on A. Find the minimum and maximum value of F which can be applied so that both blocks can move combinely without any relative motion between them.



(A) $10\ \text{N}, 50\ \text{N}$

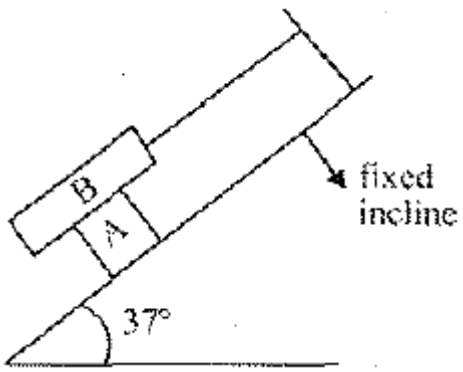
(B) $12\ \text{N}, 50\ \text{N}$

(C) $12\ \text{N}, 75\ \text{N}$

(D) None of these

Q.11

Two blocks A and B each of mass m are placed one over another on an incline as shown in Fig. When the system is released from rest the block A slides down with constant velocity while block B rests on top of A. If the coefficient of friction between A and B and between A and incline are same, then value of coefficient of friction would be



(A) $\frac{1}{4}$

(B) $\frac{3}{5}$

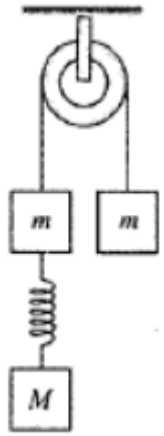
(C) $\frac{4}{5}$

(D) Information insufficient

Q.12

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The system shown in Fig. is released from rest. The spring gets elongated



(A) if $M > m$

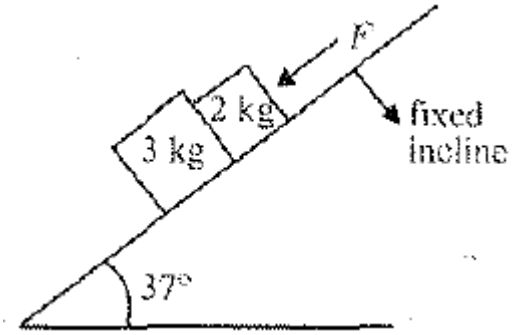
(B) if $M > 2m$

(C) if $M > m/2$

(D) For any value of M (Neglect friction and masses of pulley, string and spring)

Q.13

Two blocks of masses $3\ \text{kg}$ and $2\ \text{kg}$ are placed side by side on an incline as shown in the Fig. A force, $F = 20\ \text{N}$ is acting on $2\ \text{kg}$ block along the incline. The coefficient of friction between the block and the incline is same and equal to 0.1, find the normal contact force exerted by $2\ \text{kg}$ block on $3\ \text{kg}$ block.



(A) $18\ \text{N}$

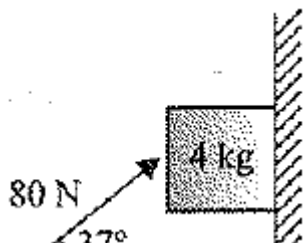
(B) $30\ \text{N}$

(C) $12\ \text{N}$

(D) $27.6\ \text{N}$

Q.14

A block of mass $4\ \text{kg}$ is pressed against the wall by a force of $80\ \text{N}$ as shown in the Fig. Determine the value of friction force and block's acceleration, (Take $\mu_s = 0.2,\ \mu_k = 0.15$)



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(A) $8\ \text{N}, 0\ \text{m/s}^2$

(B) $32\ \text{N}, 6\ \text{m/s}^2$

(C) $8\ \text{N}, 6\ \text{m/s}^2$

(D) $32\ \text{N}, 2\ \text{m/s}^2$

Q.15

Two persons are holding a rope of negligible weight tightly at its ends so that it is horizontal. A $15\ \text{kg}$ weight is attached to the rope at the mid point which now no longer remains horizontal. The maximum tension required to completely straighten the rope is

(A) $15\ \text{kg}$

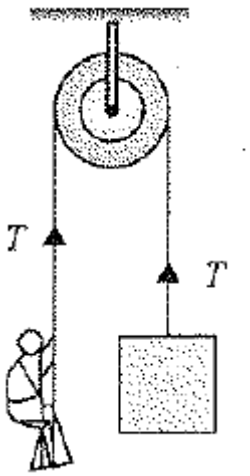
(B) $15/2\ \text{kg}$

(C) $5\ \text{kg}$

(D) Infinitely large

Q.16

In order to raise a mass of $100\ \text{kg}$ a man of mass $60\ \text{kg}$ fastens a rope to it and passes the rope over a smooth pulley. He climbs the rope with acceleration $5\ \text{g}/4$ relative to the rope (sec Fig.). The tension in the rope is (take $g = 10\ \text{m/s}^2$)



(A) $1432\ \text{N}$

(B) $928\ \text{N}$

(C) $1219\ \text{N}$

(D) $642\ \text{N}$

Q.17

A block of metal weighing $2\ \text{kg}$ is resting on a frictionless plane. It is struck by a jet releasing water at a rate of $1\ \text{kg/s}$ and a speed of $5\ \text{m/s}$. The initial acceleration of the block will be

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(A) $2.5\ \mathrm{m/s^2}$

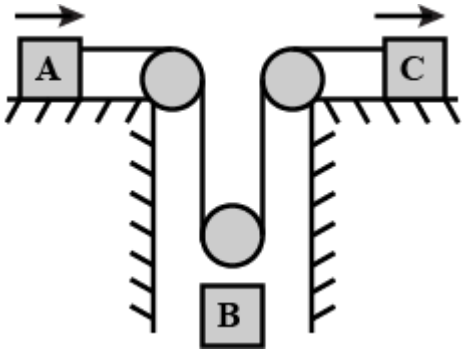
(B) $5\ \mathrm{m/s^2}$

(C) $10\ \mathrm{m/s^2}$

(D) $20\ \mathrm{m/s^2}$

Q.18

Blocks A and C start from rest and move to the right with acceleration $a_A = 12t\ \mathrm{m/s^2}$ and $a_C = 3\ \mathrm{m/s^2}$. Here t is in seconds. The time when block B again comes to rest is



(A) $2\ \mathrm{s}$

(B) $1\ \mathrm{s}$

(C) $3/2\ \mathrm{s}$

(D) $1/2\ \mathrm{s}$

Q.19

Mark out the most appropriate statement.

(A) The normal force is the same thing as the weight.

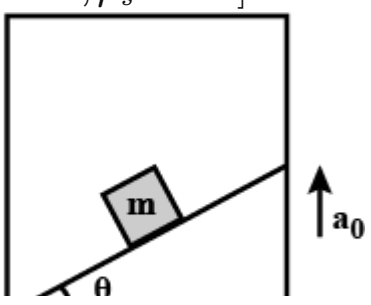
(B) The normal force is different from the weight, but always has the same magnitude.

(C) The normal force is different from the weight, but the two form an action-reaction pair according to the Newton's third law.

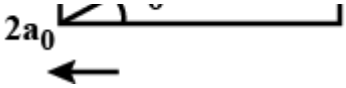
(D) The normal force is different from the weight, but the two may have same magnitude in certain cases.

Q.20

For the situation, the block is stationary w.r.t. incline fixed in an elevator. The elevator has an acceleration of $\sqrt{5}a_0$ whose components are shown in the figure. The surface is rough and coefficient of static friction between the incline and block is μ_s. Determine the magnitude of force exerted by incline on the block. Take $a_0 = \frac{g}{2}$ and $\theta = 37^\circ, \mu_s = 0.6]$



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(A) $\frac{mg}{10}$

(B) $\frac{9mg}{25}$

(C) $\frac{3mg}{25} \times \sqrt{41}$

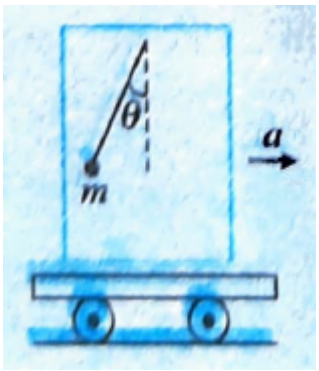
(D) $\frac{\sqrt{13}mg}{2}$

Q.21

A disc of mass 10 g is kept floating horizontally by throwing 10 marbles s^{-1} against it from below. If the mass of each marble is 5 g, calculate the velocity with which the marbles are striking the disc. Assume that the marbles strike the disc normally and rebound downward with the same speed.

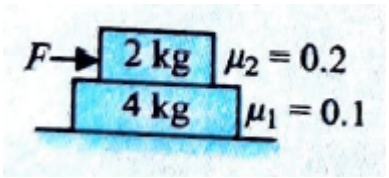
Q.22

A bob of mass $m = 50$ g is suspended from the ceiling of a trolley by a light inextensible string. If the trolley accelerates horizontally, the string makes an angle $\theta = 37^\circ$ with the vertical. Find the acceleration of the trolley. (mark answer to nearest integer) Take the value of g as $10\ \mathrm{m/s^2}$



Q.23

In the situation shown in the figure,



What minimum force F will make any part or whole system move?

Q.24

A man of mass m is moving with a constant acceleration a w.r.t. plank. The plank lies on a smooth horizontal floor. If the mass of plank is also m, calculate the acceleration of plank and man w.r.t. ground, and frictional force extended by plank on man. (given value of accleration as 2)

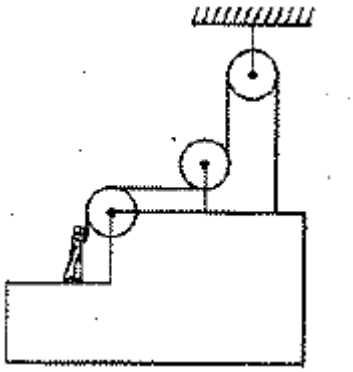


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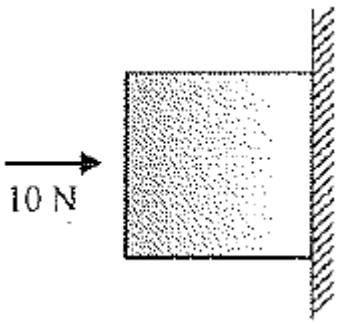
Q.25

A system is shown in the Fig. A man standing on the block is pulling the rope. Velocity of the point of string in contact with the hand of the man is $2\ \mathrm{m/s}$ downwards. The velocity of the block will be [assume that the block does not rotate]



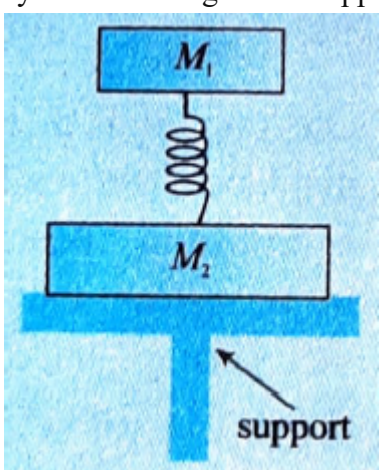
Q.26

A horizontal force of $10\ \mathrm{N}$ is necessary to just hold a block stationary against a wall. The coefficient of friction between the block and the wall is 0.2 (Fig.). The weight of the block is



Q.27

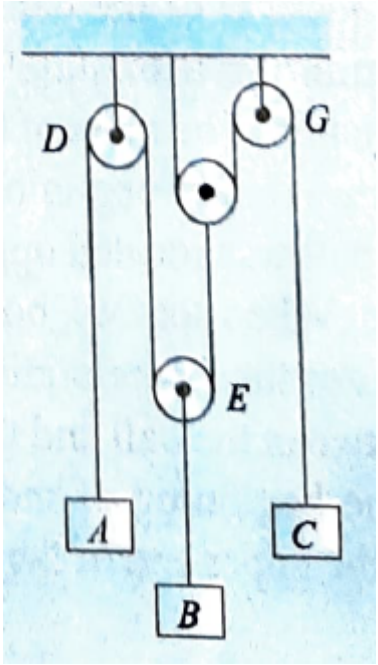
The system of two weights with masses M_1 and M_2 are connected with weightless spring as shown in figure. The system is resting on the support S. Find the acceleration of M_1 weight just after the support S is quickly removed.



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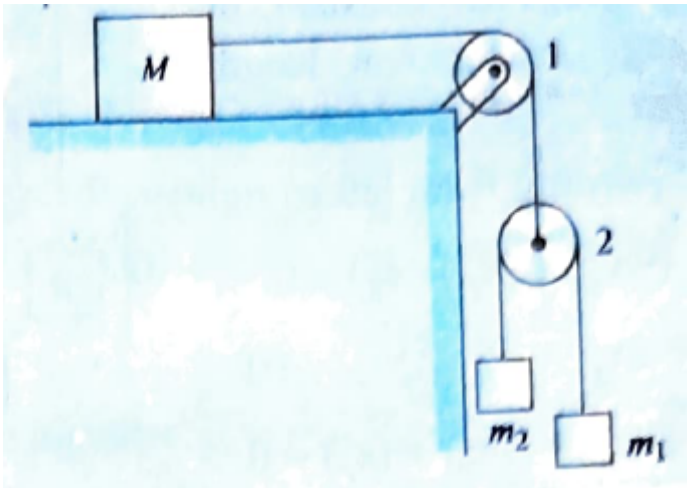
Q.28

Pulleys shown in the system are massless and frictionless. Threads are inextensible. The mass of blocks A, B, and C are $m_1 = 2\ \text{kg}, m_2 = 4\ \text{kg}$, and $m_3 = 2.75\ \text{kg}$, respectively. Calculate the acceleration of C block.



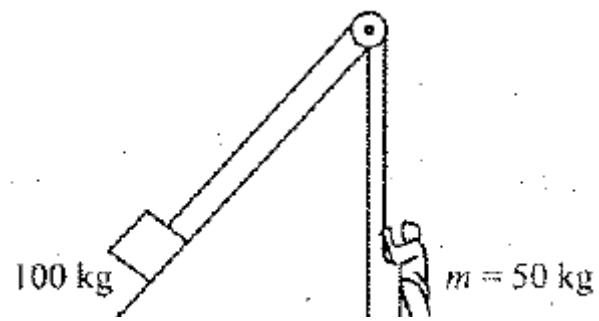
Q.29

In the arrangement shown in figure, $m_1 = 1\ \text{kg}, m_2 = 2\ \text{kg}$. Pulleys are massless and strings are light. For what value of M, the mass m_1 moves with constant velocity.



Q.30

In the given Fig., by what acceleration the boy must go up so that $100\ \text{kg}$ block remains stationary on the wedge. The wedge is fixed and friction is absent everywhere (take $g = 10\ \text{m/s}^2$)



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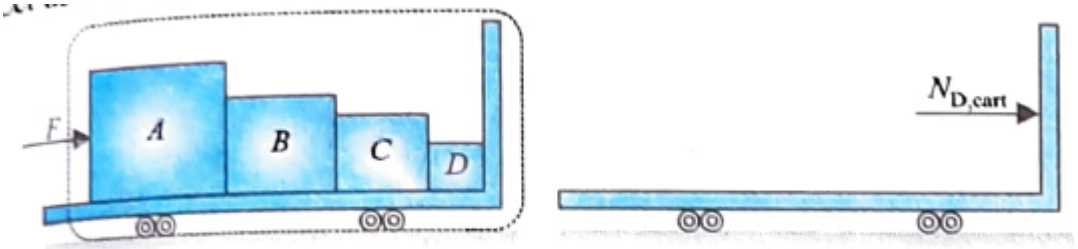
Answers & Solutions

Q.1 Answer:

24

Solution:

Let us take the cart and the blocks together as system.



System: 'cart+blocks' F.B.D. of system: 'cart'

Acceleration of the system,

$a = \frac{F}{m_A+m_B+m_C+m_D+m_{car}} = \frac{60}{2+2+1+1+4} = 6\ \text{m/s}^2$

The cart moves because of the normal reaction applied by the block ' D ' on the cart. Now considering the F.B.D. of the cart (considering horizontal forces only)

$\therefore \quad N_{D,can} = m_{car} \cdot a = 4 \times 6 = 24\ \text{N}$

Q.2 Answer:

26

Solution:

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The velocity of the sphere is same as that of the cube, which is given as $\vec{v} = 5t\hat{i} + 2t\hat{j}$

Hence, acceleration of the sphere: $\vec{a} = \frac{d\vec{v}}{dt}$

or $\vec{a} = (5\hat{i} + 2\hat{j})\text{ms}^{-2}$

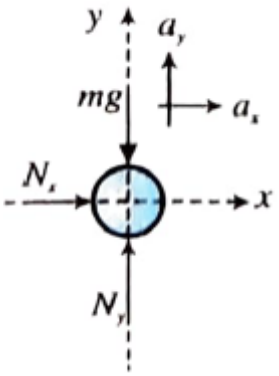
Hence, $a_x = 5\ \text{ms}^{-2}$ and $a_y = 2\ \text{ms}^{-2}$

From FBD of sphere,

$N_x = ma_x = 2 \times 5 = 10\ \text{N}$

$N_v - mg = ma_y \Rightarrow N_v = 2 \times 10 + 2 \times 2 = 24\ \text{N}$

Total force $= \sqrt{N_x^2 + N_y^2} = \sqrt{(10)^2 + (24)^2} = 26\ \text{N}$



Q.3 Answer:

$M/2$

Solution:

The spring will exert maximum force when the ball is at its lowest position. If the ball has descended through a distance x to reach the position.

$mgx = \frac{1}{2}Kx^2$ or $x = 2\text{mg}/\text{K}$ **(i)**

For the block B to leave contact spring force

$Kx = Mg$ **(ii)**

Comparing equations (i) and (ii), $m = M/2$

Q.4 Answer:

$\frac{R_3}{m}$

Solution:

Three forces are acting on a particle of mass m initially in equilibrium. If the first two forces R_1 and R_2 perpendicular to each other, R_3 has to balance the resultant of the other two forces. So,

$\text{R}_3 = \sqrt{(\text{R}_1)^2 + (\text{R}_2)^2}$

Now, the force R_3 is removed

So, the particle moves with the resultant force R_1 and R_2

Now, the acceleration is

$\text{R}_3 = \text{ma}$

$\text{a} = \frac{\text{R}_3}{\text{m}}$

Hence, the acceleration of the particle is $\frac{\text{R}_3}{\text{m}}$.

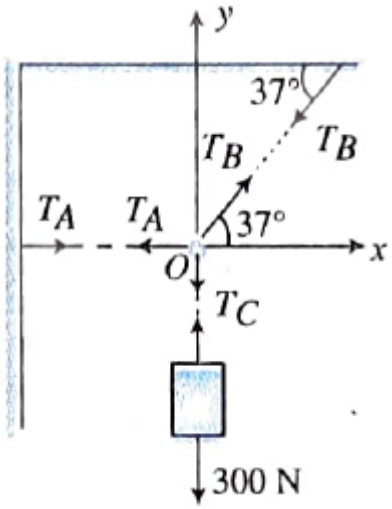
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Q.5 Answer:

$\mathrm{Ta} = 400\ \mathrm{N},\ \mathrm{Tb} = 500\ \mathrm{N}, \mathrm{Tc} = 300\ \mathrm{N}$

Solution:

Method I: Considering equilibrium of each part of system The whole system is in equilibrium; therefore, for each part $\sum \vec{F} = 0$. From the free-body diagram of block $C, T_C = 300$ N.



Now consider the equilibrium of point O

$$\sum F_x = 0 \text{ or } T_B \cos 37^\circ - T_A = 0$$

$$\therefore \quad T_A = T_B \cos 37^\circ = T_B \cdot \tfrac{4}{5} \quad ..(\mathrm{i})$$

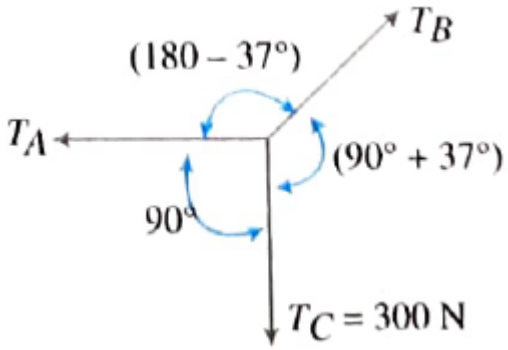
and $\sum F_y = 0$ **or** $T_B \sin 37^\circ - T_C = 0$ **...(ii)**

$$\therefore \quad T_B = \frac{T_C}{\sin 37^\circ} = \frac{300}{3/5} = 500\ \mathrm{N}$$

From Eq. (i), we get

$$T_A = \tfrac{4}{5} T_B = \tfrac{4}{5} \times 500 = 400\ \mathrm{N}$$

Method II: Using Lami's theorem



By Lami's theorem, we have

$$\frac{T_A}{\sin(90+37^\circ)} = \frac{T_B}{\sin 90^\circ} = \frac{T_C}{\sin(180-37^\circ)}$$

But $TC = 300$ N

$$\text{and } T_B = \frac{T_C}{\sin 37^\circ} = \frac{300}{3/5} = 500\ \mathrm{N}$$

$$T_A = T_C \left(\frac{\sin(90+37^\circ)}{\sin(180-37^\circ)}\right) = 300 \left(\frac{\cos 37^\circ}{\sin 37^\circ}\right) = 400\ \mathrm{N}$$

Q.6 Answer:

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5.36 N

Solution:

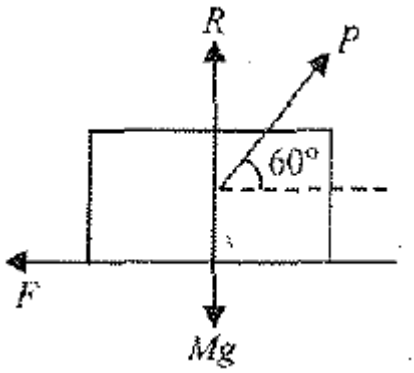
$R + P\sin 60 = Mg$

$R = Mg - P\sin 60^\circ$

Frictional force

$F = \mu R$

$= \mu\left[Mg - P\sin 60^\circ\right.$



The body just moves when

$P\cos 60^\circ = F$

Or $P\cos 60^\circ = \mu\left[Mg - P\sin 60^\circ\right]$ **or** $\frac{P}{2} = \frac{1}{2}\left[1\times 10 - \frac{P\sqrt{3}}{2}\right]$

Or $P = 5.36$ N

Q.7 Answer:

$\mu Mg/F_0$

Solution:

$f_l = \mu Mg, f = F = F_0 t$

Motion will start when $f = f_l$

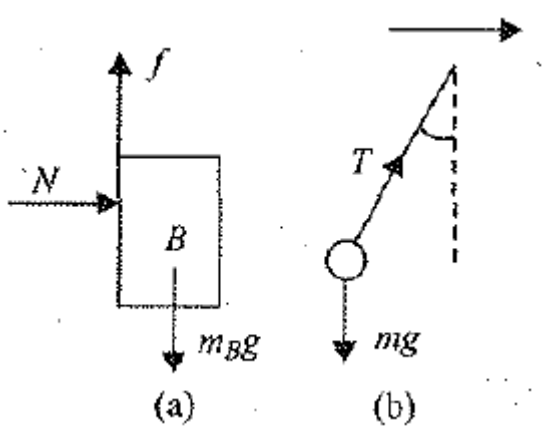
$\Rightarrow \quad F_0 T = \mu Mg \quad \Rightarrow T = \frac{\mu Mg}{F_0}$

Q.8 Answer:

$\tan^{-1}(2)$

Solution:

In the free body diagram of B **[Fig.]**



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$$N = m_B a \quad \text{(i)}$$
$$f = m_B g$$
$$\mu N = m_B g \quad \text{(ii)}$$

From equations (i) and (ii), $a = \frac{g}{\mu} = 20\ \text{m/s}^2$

In the free body diagram of the bob [Fig. $7.608(\text{ b})$**]**

$$T \sin\theta = ma \quad \text{(iii)}$$
$$T \cos\theta = mg \quad \text{(iv)}$$

From equations (iii) and (iv), we get

$$\tan\theta = \frac{a}{g} \quad \Rightarrow \quad \theta = \tan^{-1}(2)$$

Q.9 Answer:

$3/4$

Solution:

For a frictionless surface,

$$a = g \sin 45^\circ = g/\sqrt{2}$$
$$\therefore \quad l = \tfrac{1}{2} \times \tfrac{g}{\sqrt{2}} \times t_2^2 \text{ (1)}$$

In the presence of friction,

$$a = \frac{g}{\sqrt{2}} - \frac{\mu g}{\sqrt{2}} = \frac{g}{\sqrt{2}}(1-\mu)$$
$$\therefore \quad l = \frac{1}{2} \times \left[\frac{g}{\sqrt{2}}(1-\mu)\right] \times t_1^2 \quad (2)$$

Dividing eq. (2) by eq. (1), we get

$$1 = (1-\mu)\frac{t_1^2}{t_2^2}, \therefore \mu = 3/4$$

Q.10 Answer:

$12\ \text{N}, 75\ \text{N}$

Solution:

$$F - N \sin 37^\circ = 6a \Rightarrow F - \tfrac{3N}{5} = 6a \ldots \text{ (i)}$$
$$N_2 = 4g - N \cos 37^\circ = 40 - \tfrac{4N}{5} \ldots \text{ (ii)}$$
$$N \sin 37^\circ - f = 4a \cdots \text{ (ii)}$$
$$\text{From eqs (i) and (ii) } F - f = 10a$$
$$\Rightarrow F - mN_2 = 10a \ldots \text{ (iv)}$$
$$\Rightarrow F - m\left[40 - \tfrac{4N}{5}\right] = 10a \ldots \text{ (v)}$$

Put the value of N **from eqs (i) and (V) and abo**

Put the value of m **to get** $a = \frac{5F-60}{42}$

$\Rightarrow$ **Now to start motion** $a > 0 \Rightarrow F > 12\ \text{N}$

Now the minimum Force F to Just start the motion is $12\ \text{N}$

Now From eqs (i) and (iv) weget $75\ \text{N}$ **(by putting** $N_2 \cdot 0$ **)**

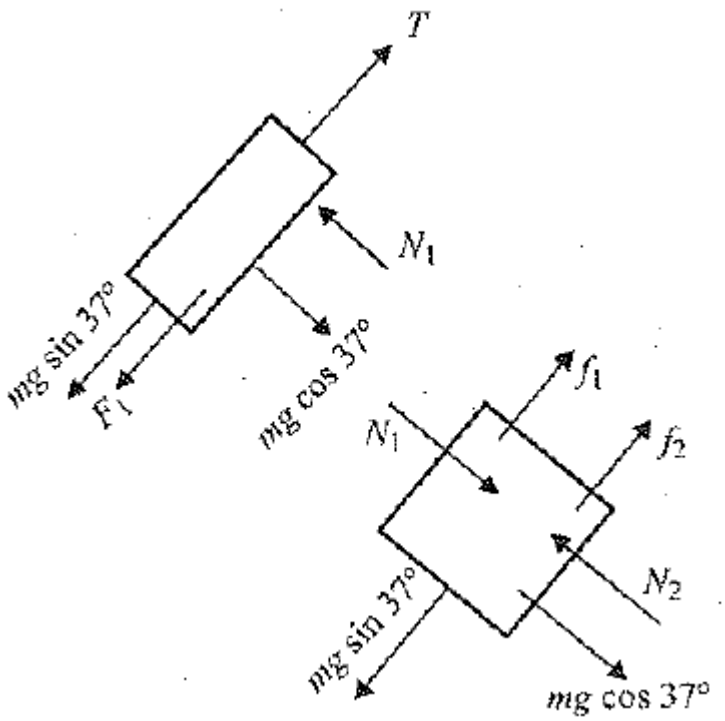
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Q.11 Answer:

$\frac{1}{4}$

Solution:

The free body diagrams of two blocks are as shown in Fig,



From free body diagram of B

$N_1 = mg\cos 37^\circ \; T = mg\sin 37^\circ + f_1 \; F_l = \mu \mathrm{N}_1$ **(where** μ **is coefficient of friction)**

From the free body diagram of A

$$N_2 = N_1 + mg\cos 37^\circ = 2mg\cos 37^\circ$$

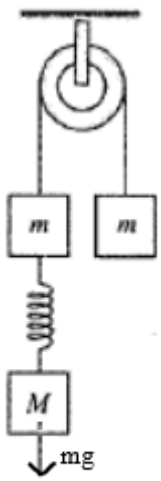
$$mg\sin 37^\circ = f_1 + f_2$$

$$f_2 = \mu N_2 \Rightarrow \mu = \frac{1}{4}$$

Q.12 Answer:

For any value of M (Neglect friction and masses of pulley, string and spring)

Solution:



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$F = kx$
$x \propto F$
$T \to M \propto x$
$n > 0 \quad T > 0$
for any value of M (M should be greater than zero.

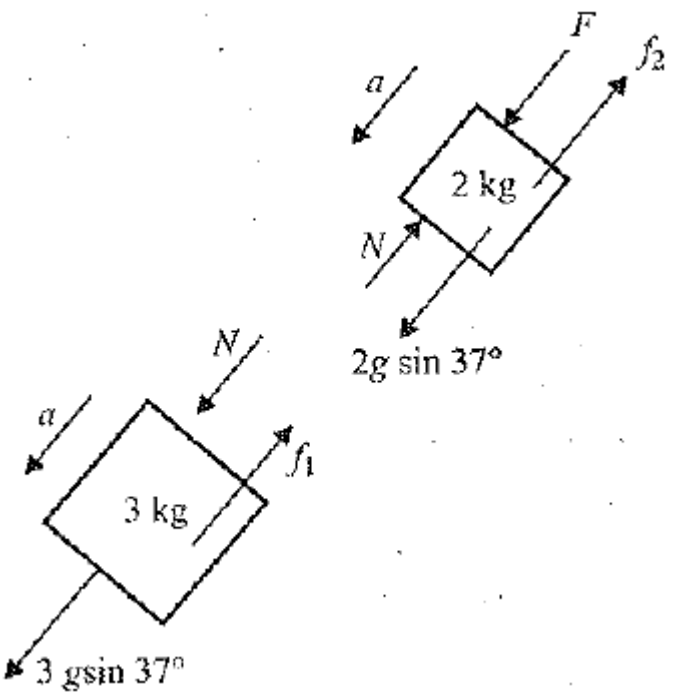
Q.13 Answer:

12 N

Solution:

The free body diagrams of two blocks are shown in Fig. Under the assumption that both the blocks are moving together,

$F + 2g \sin 37^\circ + 3g \sin 37^\circ - f_1 - f_2 = 5a$
where $f_1 = \mu \times 3g \cos 37^\circ$
and $f_2 = \mu \times 2g \cos 37^\circ$



$\Rightarrow a = \frac{46}{5}\ \mathrm{m/s^2}$
For 3 kg block, $N + 3g \sin 37^\circ - f_1 = 3a \Rightarrow N = 12\ \mathrm{N}$

Q.14 Answer:

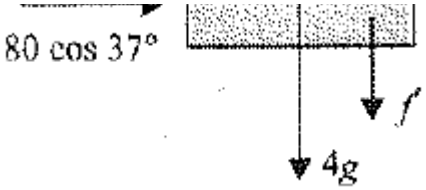
$8\ \mathrm{N}, 0\ \mathrm{m/s^2}$

Solution:

The FBD of the block is shown in the Fig. $7.638\ N = 80 \cos 37^\circ = 64\ \mathrm{N}$



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So, $f_L = 0.2 \times 64 = 32$ N

As $4g < 80 \sin 37°$**, so the frictional force will act downwards.**

Net applied force in upward direction (excluding friction force) is

$80 \sin 37° - 40 = 48 - -40 = 8$ N

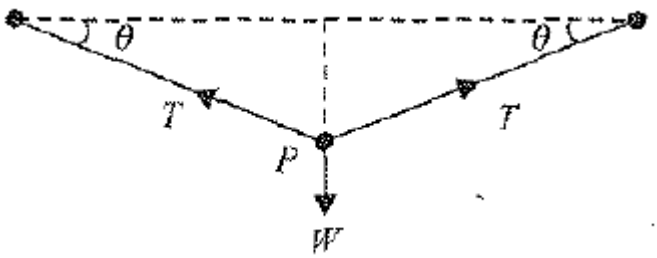
As F_{applied} **in vertical direction is** $< f_L$ **so block won't move in vertical direction and value of static friction force is,** $f = 8$ N

Q.15 Answer:

Infinitely large

Solution:

When a string is fixed horizontally as shown in Fig. (by clamping its free ends) and loaded at the middle, then for the equilibrium of point P



$2T \sin \theta = W$

$T = \frac{W}{2 \sin \theta}$

Tension in the string will be maximum when $\sin \theta$ **is minimum, i.e.,** $\theta = 0°$ **or** $\sin \theta = 0$ **and then** $T = \infty$**. However, as every string can bear a maximum finite tension lesser than breaking strength). So this situation cannot be realized practically. We conclude that a string can never remain horizontal when loaded at the middle howsoever great mass be the tension applied.**

Q.16 Answer:

1219 N

Solution:

Let T **be the tension in the rope and a the acceleration of rope. The absolute acceleration of the man is therefore** $\left(\frac{5g}{4} - a\right)$**. Equations of motion for mass and man give**

$T - 100g = 100a \quad \text{(i)}$

$T - 60g = 60\left(\frac{5g}{4} - a\right) \quad \text{(ii)}$

Solving equations (i) and (ii), we get,

$T = 1218.75 \text{ N} \approx 1219 \text{ N}$

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Q.17 Answer:

$2.5\ \mathrm{m/s^2}$

Solution:

The water jet striking the block at the rate of $1\ \mathrm{kg/s}$ at a speed of $5\ \mathrm{m/s}$ will exert a force on the blank

$F = v\frac{dm}{dt} = 5 \times 1 = 5\ \mathrm{N}$

Under the action of this force of $5\ \mathrm{N}$, the block of mass $2\ \mathrm{kg}$ will move with an acceleration given by

$a = \frac{F}{m} = \frac{5}{2} = 2.5\ \mathrm{m/s^2}$

Q.18 Answer:

$1/2\ \mathrm{s}$

Solution:

From constraint relations, we can see that the acceleration of block B in upward direction is

$a_B = \left(\frac{a_C + a_A}{2}\right)$ with proper signs.

So, $a_B = \left(\frac{3-12t}{2}\right) = 1.5 - 6t$

or $\frac{dv_B}{dt} = 1.5 - 6t$

$\int_0^{v_D} dv_B = \int_0^1 (1.5 - 6t)dt$

or or $v_B = 1.5t - 3t^2$

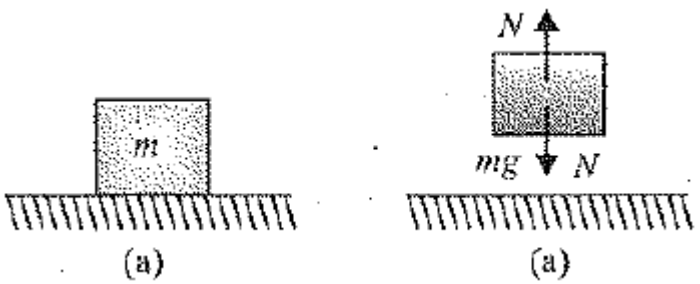
or $v_B = 0$ at $t = 1/2\ \mathrm{s}$

Q.19 Answer:

The normal force is different from the weight, but the two may have same magnitude in certain cases.

Solution:

If we consider the situation shown (a) in Fig. the FBD would be as shown in Fig.



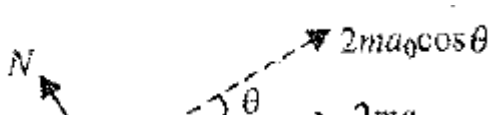
In this situation, the option (b) seems to be correct, but in other case like if this system is in a life moving with some acceleration, then $N \neq mg$. So, (d) is the correct option.

Q.20 Answer:

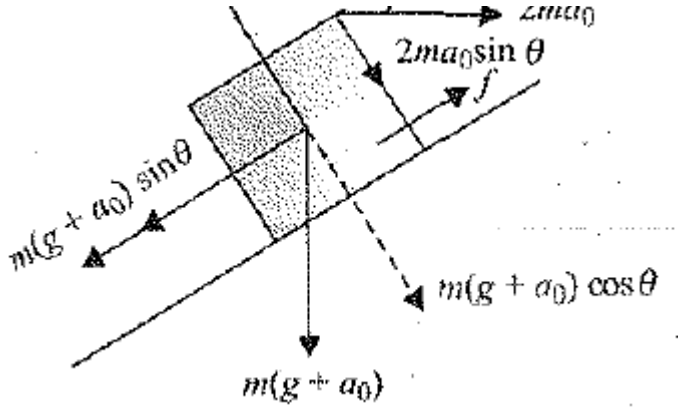
$\frac{\sqrt{13}mg}{2}$

Solution:

The FBD of block from the lift frame is shown in Fig. From the given data, as $m(g + a_0)\sin\theta > 2ma_0\cos\theta$.



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So, the friction force acts upwards.

$$f = m(g+a_0)\sin\theta - 2ma_0\cos\theta$$
$$= \frac{9g}{10} - \frac{4mg}{5} = \frac{mg}{10}$$
$$N = m(g+a_0)\cos\theta - 2ma_0\sin\theta = \frac{9mg}{5}$$

As $f_L = \mu_s N = \frac{18mg}{50} = \frac{9mg}{25} > f$ **so the static friction.**

Reaction force,

$$R = \sqrt{f^2+N^2} = \frac{mg}{5}\sqrt{\frac{1}{4}+9^2} = \frac{mg\sqrt{13}}{2}$$

Q.21 Answer:

—

Solution:

Given that:

Mass of disc, $\mathrm{M} = 10\ \mathrm{g} = 0.010\ \mathrm{kg}$

Mass of each marble, $\mathrm{m} = 5\ \mathrm{g} = 0.005\ \mathrm{kg}$

Suppose the number of marbles striking the disc per second=n and velocity with which marbles strike disc is v**.**

Then Weight of disc acting downward= Mg

Change in velocity of marbles when they rebound $= \mathrm{v} - (-\mathrm{v}) = 2\mathrm{v}$

Therefore the change in momentum of each marble when it strikes the disc= $\mathrm{m} * 2\mathrm{v} = 2\mathrm{mv}$

and total momentum imparted per second to the disc $= 2\mathrm{mnv}$ **= The force exerted by the marbles in the upward direction The disc will remain at rest if the net force acting on it is zero provided it is initially at rest.**

Therefore for the disc to remain at rest,

Weight of disc=Upward force on it

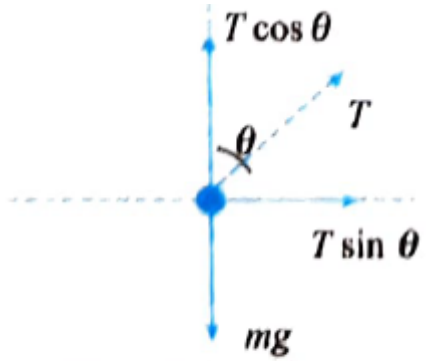
$$= \mathrm{Mg} = 2\mathrm{mnv}$$
$$\mathrm{v} = \frac{\mathrm{Mg}}{2\mathrm{mn}} = \frac{(0.010*9.8)}{(2*0.005*10)}$$
$$= 0.98\ \mathrm{m/s}$$

Q.22 Answer:

—

Solution:

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FBD of the bob w.r.t observer standing on ground

Method 1: Problem solving from ground frame. The obscrver on the ground will observe the bob moving with acceleration a same as trolley. In FBD. he will observe only two real forces, weight and tension force. The bob is at cquilibrium in vertical dircction, hence writing equation of motion in vertical direction

$T\cos(\theta$ lilg **...(i)**

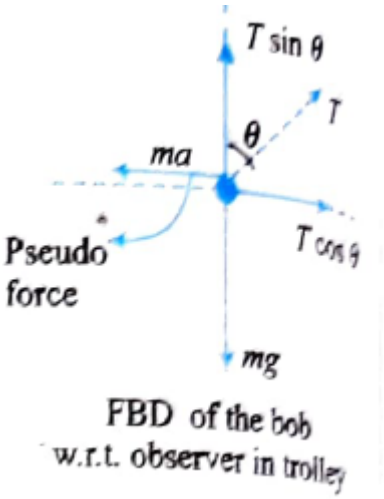
Along horizontal direction, the bob is accelerating with, acceleration a

$T\sin\boldsymbol{\theta} = ma$

Using (i) and (ii), $\tan\theta = a/g$

$\therefore \quad a = g\tan\theta = 10\tan 37^\circ = 7.5\ \mathrm{m\ s^{-2}}$

Therefore, acceleration of the trolley is $7.5\ \mathrm{m\ s^{-2}}$



FBD of the bob w.r.t. observer in trolley

Method 2: Problem solving from noninertial frame. If an observer sitting in the trolley and observes the bob, he will observe the bob at rest. If he asks to draw free-body diagram of bob he will apply pseudo force apart from real forces acting on the bob.

The bob is at equilibrium in vertical direction, hence writing equation of motion in vertical direction

$T\cos\theta = mg$ **(iii)**

Along horizontal direction the bob is at rest.

$T\sin\theta = ma$ **(iv)**

Using (iii) and (iv) $\tan\theta = a/g$

$\therefore \quad a = g\tan\theta = 10\tan 37^\circ = 10\tan 37 = 10\times 0.75 = 7.5.$ **nearest integer to** 7.5 **is 8.**

Q.23 Answer:

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Solution:

Between 4 kg and ground:

$f_{l_1} = f_{k_1} = \mu_1(2+4)g = 6$ N

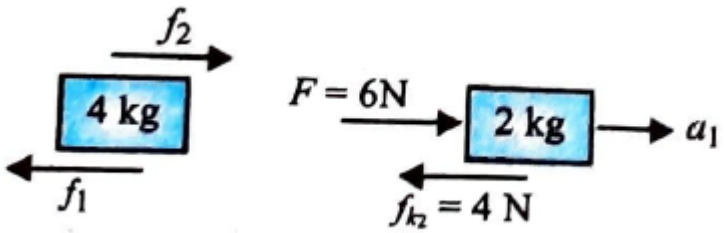
Between 2 kg and 4 kg :

$f_{l_2} = f_{k_2} = \mu_2 2g = 4$ N

(a) Limiting friction between 2 kg and 4 kg is less than that between 4 kg and ground. So 2 kg will slip over 4 kg if

$F > f_{l_2} = 4N \Rightarrow F_{\min} = 4$ N

(b) Let us consider the FBD of 4 kg : Acceleration to 4 kg block will be provided by f_2. But the maximum value of f_2 is less than the maximum value of f_1. Hence, 4 kg will not accelerate. So $a_4 = 0$



For 2 kg : $a_1 = \frac{6-4}{2} = 1$ m s^{-2}

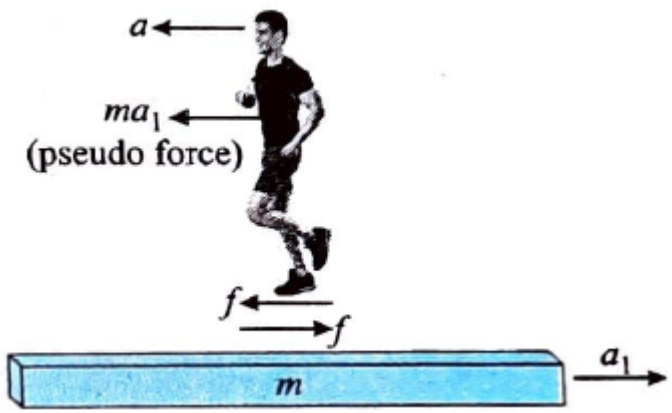
Q.24 Answer:

—

Solution:

Let us consider the acceleration of the plank be a_1 (rightwards). Now considering the motion of the man w.r.t. plank.

Equation of motion for man: $ma_1 + f = ma$ **...(i)**



Equation of motion for plank: $f = ma_1$ **...(ii)**

From (i) and (ii): $ma_1 + ma_1 = ma \Rightarrow 2ma_1 = ma \Rightarrow a_1 = \frac{a}{2}$

Frictional force between man and plank

From (i) $f = ma_1 = \frac{ma}{2}$

Acceleration of man $\vec{a}_{\text{man}} = \vec{a}_{\text{man,plank}} + \vec{a}_{\text{plank}} = a + \left(-\frac{a}{2}\right) = \frac{a}{2}$

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(Towards left)

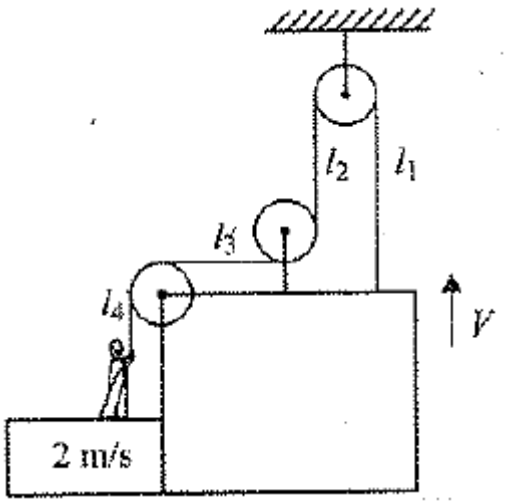
Q.25 Answer:

—

Solution:

As in Fig.

$l_1 + l_2 + l_3 = C$



$\frac{dl_1}{dt} + \frac{dl_2}{dt} + \frac{dl_3}{dt} + \frac{dl_4}{dt} = 0$

$-v - v + 0 + v + 2 = 0$

$v = 2 \text{ m/s}$

Q.26 Answer:

—

Solution:

Normal reaction on the block from the wall:

$R = F = 10 \text{ N}$

Weight of the block will be balanced by the frictional force

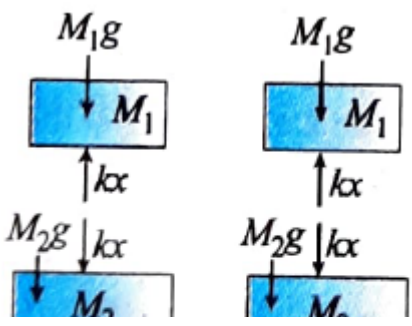
$W = \text{ Fictional force } = \mu R = 0.2 \times 10 = 2 \text{ N}$

Q.27 Answer:

—

Solution:

The force of spring does not change instantaneously, so find spring force at initial instant.



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FBD initially FBD when support is removed

Initially, $M_1 g = kx$

When support is removed, spring force does not change.

For M_1 : $M_1 g - kx = M_1 a_1$ **or** $a_1 = 0$

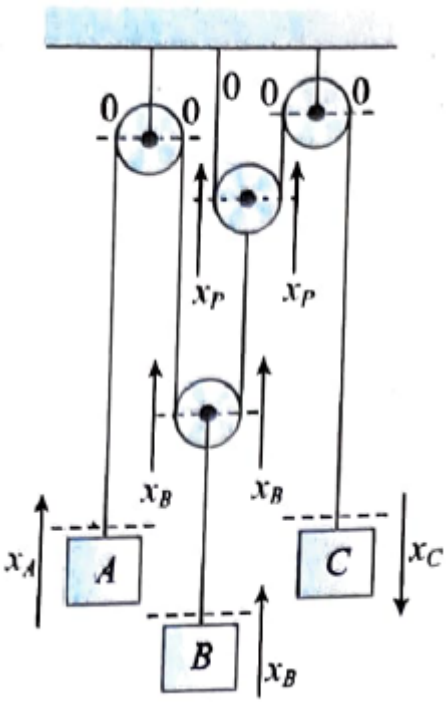
For M_2 : $M_2 g + kx = M_2 a_2$ **or** $a_2 = \frac{(M_1+M_2)g}{M_2}$

Q.28 Answer:

—

Solution:

Let the acceleration of blocks $\mathcal{A}$ **and** B **be** a **and** b **vertically upwards, and acceleration of block** C **is** c **downward, respectively.**



As length of strings are constant, the sum of change in segment lengths should be zero.

For string 1 : [writing from left to right]

$(-x_A) + (-x_B) + (x_P - x_B) = 0$

$x_P = x_A + 2x_B \quad \ldots(\text{i})$

For string 2 : $(-x_p) + (-x_P) + (x_C) = 0$

$x_C = 2x_p \quad \ldots(\text{ii})$

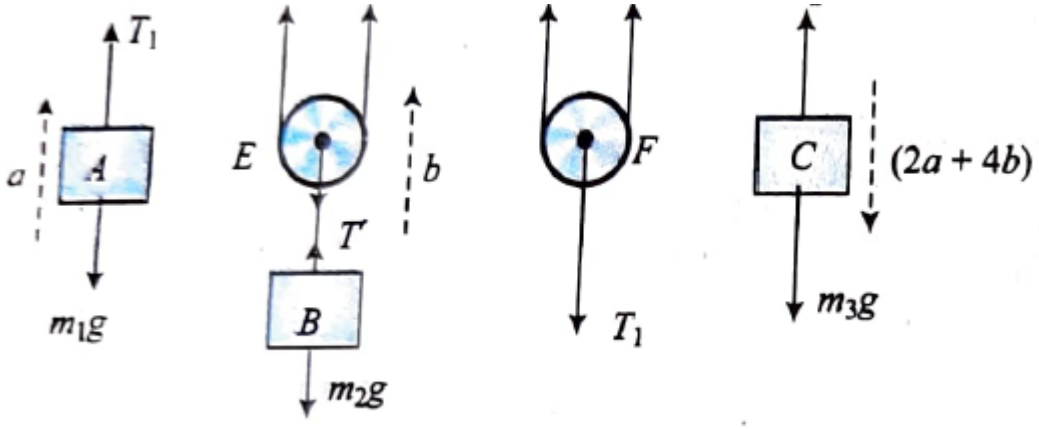
From (i) and (ii): $x_C = 2x_A + 4x_B$

$\Rightarrow \quad c = 2a + 4b \quad \ldots(\text{iii})$

Now considering FBDs, we get the following:

$T_1 \quad T_1 \qquad T_2 \quad T_2 \qquad T_2$

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For block A, $T_1 - m_1 g = m_1 a$ **...(i)**

For block B, $2T_1 - m_2 g = m_2 b$ **...(ii)**

For pulley F, $T_1 = 2T_2$ ...(iii)

For block C, $m_3 g - T_2 = m_3(2a + 4b)$ ...(iv) **For block C,** $m_3 g - T$ **Solving above equations,**

Solving above equations,

$$T_1 = 22 \text{ N}, T_2 = 11 \text{ N}$$

$a = 1 \text{ m s}^{-2}, b = 1 \text{ m s}^{-2}$

Hence, acceleration of block A,

$a = 1 \text{ m s}^{-2}(\uparrow)$

Acceleration of block B

$b = 1 \text{ m s}^{-2}(\uparrow)$

Acceleration of block $C = (2a + 4b) = 6 \text{ ms}^{-2}(\downarrow)$

Q.29 Answer:

—

Solution:

Mass m_1 moves with constant velocity if tension in the lower string,

$T_1 = m_1 g = (1)(10) = 10 \text{ N}$ **...(i)**

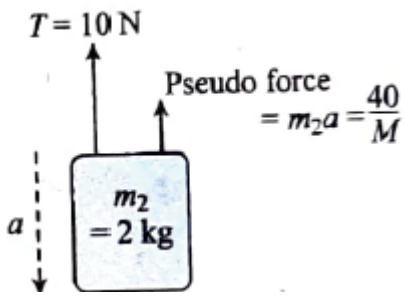
Tension in the upper string,

$T_2 = 2T_1 = 20 \text{ N}$ **...(ii)**

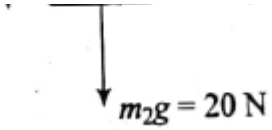
Acceleration of block M is, therefore,

$a = \frac{T_2}{M} = \frac{20}{M}$ **...(iii)**

This is also the acceleration of pulley 2.



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The absolute acceleration of mass m_1 is zero. Thus, the acceleration of m_1 relative to pulley 2 is a upward or acceleration of m_2 with respect to pulley 2 is a downward. Drawing free-body diagram of m_2 with respect to pulley 2 (figure above).

Equation of motion gives

$20 - \frac{40}{M} - 10 = 2a = \frac{40}{M}$

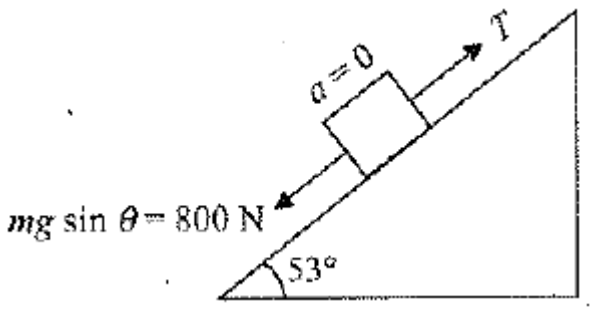
Solving this, we get $M = 8$ kg.

Q.30 Answer:

—

Solution:

For a block to be stationary $T = 800$ N **(Fig.)**



If a man moves up by acceleration a.

$$T - mg = ma$$

$$8000 - 5000 = 50a \Rightarrow a = 6 \text{ m/s}^2$$

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